

Radar_Sensor_Preprocessing Qualification Test Specification

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Chapter 1

Covered Requirements

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FTCP-8842

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FTCP-8190

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FTCP-8190

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FTCP-8190

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 ()
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 ()
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Chapter 2

Module Documentation

2.1 F360_check_det_az_conf_and_elevation_qualtest

Functions

- test group [f360_check_det_az_conf_and_elevation_qualtest](#) ()
This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off](#) ()

2.1.1 Detailed Description

2.1.2 Function Documentation

2.1.2.1 f360_check_det_az_conf_and_elevation_qualtest()

```
test group f360_check_det_az_conf_and_elevation_qualtest ( )
```

This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.1.2.2 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation ()

Purpose:\n If the confidence value is too low (3 being the lowest) or the elevation angle is suspiciously high, do not allow detection to start a new track.

Requirements FTCP-8842

Step 1:\n

Generate Azimuth confidence level and elevation angle for MRR3 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:\n

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

confidence value is too low or the elevation angle is high,new track is not started.

confidence value is lower boundary value and the elevation angle is low, new track is not started.

confidence value is lower boundary value or the elevation angle is too high,new track is not started.

Action:\n

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:\n

Check if the confidence level and elevation angle are in range and the results are as expected

2.1.2.3 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1 ()

Purpose: Generate Azimuth confidence level and elevation angle for SRR4 sensor detections (Need to modify the test case once there is requirement update)

Requirements FTCP-8842

Step 1:

Generate Azimuth confidence level and elevation angle for SRR4 sensor detections (Need to modify the test case once there is requirement update)

Precondition:

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated SRR4 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR4 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR4 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated SRR4 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated SRR4 sensor valid detections based on f_super_res and azimuth

Action:

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:

Check if the confidence level and elevation angle are in range and the results are as expected

2.1.2.4 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2 ()

Purpose:\n Generate Azimuth confidence level and elevation angle for SRR5 seonsor detections (Need to modify the test case once there is requireme

Requirements FTCP-8842

Step 1:\n

Generate Azimuth confidence level and elevation angle for SRR5 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:\n

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated SRR5 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR5 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR5 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated SRR5 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated SRR5 sensor valid detections based on f_super_res and azimuth

Action:\n

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:\n

Check if the confidence level and elevation angle are in range and the results are as expected

2.1.2.5 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3 ()

Purpose:\n Generate Azimuth confidence level and elevation angle for MRR360 seonsor detections (Need to modify the test case once there is require

Requirements FTCP-8842

Step 1:\n

Generate Azimuth confidence level and elevation angle for F360 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:\n

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated MRR360 sensor detections that are suspicious based on elevation angle

Check for unassociated MRR360 sensor detections that are suspicious based on elevation angle

Check for unassociated MRR360 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated MRR360 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated MRR360 sensor valid detections based on f_super_res and azimuth

Action:\n

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:\n

Check if the confidence level and elevation angle are in range and the results are as expected

2.1.2.6 f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off()

f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off ()

Purpose: Test that the the elevation check is not performed if the f_mrr360_filter_away_big_elev_angle calib is false

Requirements FTCP-8842

Step 1:

Initialize default parameters

Precondition:

Set host speed to less than min_host_speed so that the first check is not performed.

Create 5 detection with raw_detection_list->processed.f_ok_to_use = true

1. Sensor type = MRR3, $-6 < \text{elevation angle} < 6$ deg (0.104719755119660 rad)
2. Sensor type = MRR3, elevation angle > 6 deg (0.104719755119660 rad)
3. Sensor type = SRR5, elevation angle > 6 deg (0.104719755119660 rad)
4. Sensor type = MRR3, elevation angle < -6 deg (-0.104719755119660 rad)
5. Sensor type = SRR5, elevation angle < -6 deg (-0.104719755119660 rad)

Action:

Run Check_Detection_Azimuth_Confidence_and_Elevation

Result:

Check that all the detections are still ok to use (even the MRR360 with high elevation)

2.2 Rspp_detection_double_bounce_detector_qualtest

Functions

- test group [rspp_detection_double_bounce_detector_qualtest](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch](#) ()

2.2.1 Detailed Description

2.2.2 Function Documentation

2.2.2.1 rspp_detection_double_bounce_detector_qualtest()

```
test_group rspp_detection_double_bounce_detector_qualtest ( )
```

Basic test to verify that detection gets flagged as double bounce when expected and vice versa.

Setup:\n

Setup common data for all tests related to this test group

Teardown:\n

Nothing to teardown in this test group

2.2.2.2 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since azimuth condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.2.2.3 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since range condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.2.2.4 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since range rate condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.2.2.5 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true ( )
```

Purpose:\n Test that detection is flagged as double bounce

Requirements FTCP-8841

Result:\n

We only expect the detection with higher range to be flagged as double bounce

2.3 System_test_rspp_detection_double_bounce_detector

Functions

- test group `system_test_rspp_detection_double_bounce_detector` ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- `system_test_rspp_detection_double_bounce_detector_Double_Bounce_True` ()
- `system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True` ()
- `system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario` ()

2.3.1 Detailed Description

2.3.2 Function Documentation

2.3.2.1 system_test_rspp_detection_double_bounce_detector()

```
test_group system_test_rspp_detection_double_bounce_detector ( )
```

Basic test to verify that detection gets flagged as double bounce when expected and vice versa.

Setup:\n

Setup common data for all tests related to this test group

Setup 4 detections in a double bounce pattern. Modify these values and maybe add more detections to define test cases. The following detections are setup:

det[x]	.range	.range_rate	.azimuth	.sensor_id
det[0]	10.0	2.0	0.1	1
det[1]	20.0	4.0	0.11	1
det[2]	30.0	6.0	0.11	1
det[3]	40.0	8.0	0.11	1

Teardown:\n

Nothing to teardown in this test group

2.3.2.2 system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario()

```
system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario ( )
```

Purpose:\n Test Arbitrary Scenario 1

Requirements NA

Step 1:\n

Testing arbitrary scenario

Precondition:\n

Creating new set of 7 detections:

2.3.2.3 -----X6-----

2.3.2.4 -----O3---X4-----x5-----

2.3.2.5 -----O0-O1-O2-----

-----S0-----

Character	Type
X	Multibounce detection
O	Real detection
S	Sensor

Action:\n

Call the function

Result:\n

We only expect the detection with higher range to be flagged as double bounce

2.3.2.6 system_test_rspp_detection_double_bounce_detector_Double_Bounce_True()

system_test_rspp_detection_double_bounce_detector_Double_Bounce_True ()

Purpose:\n Simple test using two detections only, one is double bounce of the other one

Requirements NA

Step 1:

Testing 2 detections

Precondition:

Testing only 2 detections of the set

Action:

Call the function

Result:

We only expect the detection with higher range to be flagged as double bounce

2.3.2.7 system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True()

```
system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True ( )
```

Purpose: Checking multiple bounces, one reference detection have yielded 3 double bounces

Requirements NA

Step 1:

Testing multiple bounces

Precondition:

Testing 4 detections of the set

Action:

Call the function

Result:

We only expect the detection with higher range to be flagged as double bounce

2.4 F360_Compute_Detections_Properties_Based_On_Ego_Motion__↔ stopped_host

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host](#) ()

Check correctness of detection motion status determination when host does not move.

- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_low](#)
()

2.4.1 Detailed Description

2.4.2 Function Documentation

2.4.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host()

```
test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host ( )
```

Check correctness of detection motion status determination when host does not move.

Setup:\n

- Init calibs.
- Setup one detection.

2.4.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_amb_due_to_being_az_error_stat ( )
```

Purpose:

Check correctness of detection motion determination when detections is marked as f_azimuth_error_stat_mov

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as ambiguous

2.4.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_amb_due_to_low_conf_az ( )
```

Purpose:

Check correctness of detection motion determination when det's conf az is low

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as ambiguous

2.4.2.4 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_1 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to 0.0F

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.4.2.5 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_2 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to 0.0F

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.4.2.6 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_3 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is high

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.4.2.7 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_moving_dealiased_case_1 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as moving

2.4.2.8 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__↵  
__mark_low_range_rate_moving ( )
```

Purpose:

Check correctness of detection motion determination when detections is marked as f_azimuth_error_stat_mov

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as ambiguous

2.5 F360_Compute_Detections_Properties_Based_On_Ego_Motion__↔ moving_forward_host

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_1](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_2](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_3](#) ()

2.5.1 Detailed Description

2.5.2 Function Documentation

2.5.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host()

test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host ()

Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)

Setup:\n

- Init calibs.
- Setup one detection.

2.5.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_1()

f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion↔
_Classification__mark_det_as_ambiguous_dealiased_case_1 ()

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to -10.0F and relative high motion threshold

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.5.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classifi

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion↔  
_Classification__mark_det_as_ambiguous_dealiased_case_2 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to -10.0F and relative low motion threshold

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.5.2.4 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classifi

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion↔  
_Classification__mark_det_as_moving_dealiased_case_1 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190 NA.

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.6 F360_Compute_Detections_Properties_Based_On_Ego_Motion__↔ fast_moving_forward_host

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host](#) ()
Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()

2.6.1 Detailed Description

2.6.2 Function Documentation

2.6.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host()

```
test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host (
)
```

Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.

Setup:\n

- Init calibs.
- Setup one detection.

2.6.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Cla

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection__↔  
Motion_Classification__aliased_amb_det ( )
```

Purpose:\n

Check correctness of detection motion determiantion when det's range rate is too high

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.6.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification__aliased_moving_det ()

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification__aliased_moving_det ( )
```

Purpose:

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190**Precondition:**

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as moving

2.7 F360_ego_motion_compensate_dets__stopped_host_qualtest

Collaboration diagram for F360_ego_motion_compensate_dets__stopped_host_qualtest:

2.8 F360_ego_motion_compensate_dets__moving_forward_host_↔ qualtest

Collaboration diagram for F360_ego_motion_compensate_dets__moving_forward_host_qualtest:

Functions

- test group [f360_ego_motion_compensate_dets__moving_forward_host_qualtest](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_comp_calculat](#)
()

2.8.1 Detailed Description

2.8.2 Function Documentation

2.8.2.1 f360_ego_motion_compensate_dets__moving_forward_host_qualtest()

```
test_group f360_ego_motion_compensate_dets__moving_forward_host_qualtest ( )
```

Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)

Setup:\n

- Init calibs.
- Setup one detection.

2.8.2.2 f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_co

```
f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification↔  
__det_rr_comp_calculation ( )
```

Purpose:\n

Check correctness of detection range rate compensated calculation

Requirements FTCP-8402

Precondition:\n

NA

Action:\n

Call Internal_Preprocessing().

Result:\n

Detection range_rate_compensated is equal to 20.0F

Chapter 3

File Documentation

3.1 [source/unittest/rspp_check_det_az_conf_and_elevation_qualtest_↔](#) unittest.cpp File Reference

Tests

- test group [f360_check_det_az_conf_and_elevation_qualtest](#) ()
This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off](#) ()

3.1.1 Detailed Description

This test file marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle

3.2 [source/unittest/rspp_detection_double_bounce_detector_qualtest_↔](#) unittest.cpp File Reference

Tests

- test group [rspp_detection_double_bounce_detector_qualtest](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch](#) ()
- test group [system_test_rspp_detection_double_bounce_detector](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [system_test_rspp_detection_double_bounce_detector_Double_Bounce_True](#) ()
- [system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True](#) ()
- [system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario](#) ()

3.2.1 Detailed Description

This file contains basic unit tests for algorithm that flags detections as double bounce if they fit the criteria.

3.3 source/unittest/rspp_ego_motion_compensate_dets_qualtest_↔ unittest.cpp File Reference

Tests

- test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host ()`
Check correctness of detection motion status determination when host does not move.
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det_`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det_`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det_`
`()`
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`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det_`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det_`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_low_`
`()`
- test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host ()`
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma`
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 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma`
`()`
- test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host ()`
Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification`
`()`
 - `f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification`
`()`

3.3.1 Detailed Description

This file contains unit tests for content of `f360_Compute_Detections_Properties_Based_On_Ego_Motion.cpp` file

3.4 source/unittest/rspp_inputs_preprocessing_qualtest_unittest.cpp File Reference

Tests

- test group [inputs_preprocessing_qualtest](#) ()
Add brief description of test group.
- [inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements](#) ()
- test group [f360_ego_motion_compensate_dets__stopped_host_qualtest](#) ()
Check correctness of detection motion status determination when host does not move.
- [f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp](#) ()
- test group [f360_ego_motion_compensate_dets__moving_forward_host_qualtest](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_comp_calcul](#) ()

3.4.1 Detailed Description

This file contains sw qualifications tests for f360_internal_preprocessing

3.4.2 Tests

3.4.2.1 inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements()

```
inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements ( )
```

Purpose:\n Test input interface requirements.

Requirements FTCP-8256, FTCP-7799

Precondition:\n

Initialize structures of the correct type as specified in requirements.

Action:\n

Call function to test that inputs, as specified in requirements, are accepted.

Result:\n

Test should compile and run. No functionality is supposed to be tested in this test.

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